

Does Example Difficulty Transfer Across Architectures? Variance of Gradients Under Pretraining and Random Initialization

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Abstract

Data-ranking methods promise to identify which training examples matter, but it is unclear whether such rankings reflect a property of the data or of the model that produced them. We study Variance of Gradients (VoG) [Agarwal et al., 2022], a per-example difficulty score, on Imagenette (a 10-class ImageNet subset) with two architectures, ResNet-50 and ConvNeXt-Base, under two initialisation regimes (ImageNet-pretrained and random). Within a single architecture, VoG cleanly separates useful from redundant data: a pretrained ConvNeXt-Base fine-tuned on the 30% highest-VoG examples reaches 95.6% validation accuracy, *above* the full-data baseline (93.7%, +1.9), whereas the 30% lowest-VoG examples reach only 88.1% (−5.6). Our central finding concerns transfer: VoG rankings from the two architectures agree strongly when both are pretrained (Spearman $\rho = 0.56$; per-class $\rho > 0.5$ in all 10 classes) but only weakly when trained from scratch ($\rho = 0.27$; 0/10 classes). Accordingly, a pretrained ConvNeXt trained on the *ResNet*-derived ranking matches training on its own ranking (95.6 vs 95.2), while from scratch no 30% subset beats full data. Example difficulty, as captured by VoG, is not always transferable when it comes to architectures trained from scratch.

1 Introduction

Training large vision models is dominated by the cost of data: not all examples contribute equally, and a recurring hope is that we can rank examples by importance and train on the informative ones. Several scores operationalise this idea. Toneva et al. [2019] count how often an example is forgotten during training; Paul et al. [2021] rank by gradient-norm (GraNd) and error- ℓ_2 -norm (EL2N) early in training; Swayamdipta et al. [2020] characterise examples by the mean and variance of model confidence over epochs; and Sorscher et al. [2022] show that with a good difficulty metric a large fraction of data can be pruned with little loss. Variance of Gradients (VoG) [Agarwal et al., 2022] belongs to this family: it scores an example by how much the gradient of its predicted probability with respect to the input fluctuates across training, with high variance marking hard or ambiguous examples.

These methods share an implicit assumption: that a ranking computed with one model is meaningful for another. This assumption underlies *selection via proxy* [Coleman et al., 2020], where a cheap model selects data for an expensive one. Yet difficulty scores are computed *through* a specific model, so it is not obvious whether they measure a property of the data or an artifact of that model’s inductive bias and initialisation. If the former, a ranking should transfer across architectures; if the latter, it should not. To our knowledge this transfer has not been isolated for VoG, nor has the role of pretraining in it been examined.

We study VoG on Imagenette¹ with ResNet-50 [He et al., 2016] and ConvNeXt-Base [Liu et al., 2022], two architectures from different design eras, under both ImageNet-pretraining and random initialisation. Our contributions are:

- **We show that VoG selects the informative subset within an architecture.** A pretrained ConvNeXt-Base fine-tuned on the top-30% VoG examples reaches 95.6% accuracy, exceeding the full-data baseline (93.7%), while the bottom-30% reaches 88.1% (Table 2).
- **We identify pretraining as a key condition under which cross-architecture transfer of VoG emerges in this setup.** ResNet-50 and ConvNeXt-Base rankings correlate strongly when pretrained ($\rho = 0.56$, 64.6% top-30% overlap, per-class $\rho > 0.5$ in 10/10 classes) but weakly from scratch ($\rho = 0.27$, 38.9% overlap, 0/10 classes) (Table 1, Fig. 1).

¹A 10-class subset of ImageNet [Deng et al., 2009]; <https://github.com/fastai/imagenette>.

- **We show the ranking itself transfers, not just the scores.** A pretrained ConvNeXt trained on the *ResNet*-ranked high-VoG subset matches training on its own ranking (95.6 vs 95.2 fine-tuning), validating proxy-style selection across architectures.
- **We show, as a negative result, that pruning’s benefit is contingent on pretraining.** From random initialisation no 30% subset beats full data (35.9 vs 42.2), although VoG still orders informativeness correctly (high-VoG 35.9 vs low-VoG 18.0).

2 Method and Experiments

Problem setup. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be the training set with $x_i \in \mathbb{R}^{3 \times 224 \times 224}$ and $y_i \in \{1, \dots, K\}$, $K = 10$. A ranking method assigns each example a scalar score s_i ; a selection strategy keeps a subset $S \subseteq \mathcal{D}$ and trains a classifier f_θ on S . We ask (i) how validation accuracy depends on S chosen by VoG, and (ii) how far the scores s_i , and the induced ranking, agree between two architectures.

VoG score. We follow Agarwal et al. [2022]. A model is trained for $T = 5$ epochs with SGD (learning rate 0.001, momentum 0.9, weight decay 10^{-4}). After each epoch t , in **eval** mode, we compute for every example the gradient of the true-class softmax probability with respect to the input, $g_i^{(t)} = \partial p(y_i | x_i) / \partial x_i$, and average-pool it spatially to a 32×32 grid. With $\bar{g}_{i,d} = \frac{1}{T} \sum_t g_{i,d}^{(t)}$,

$$\text{VoG}_i = \frac{1}{D} \sum_{d=1}^D \sqrt{\frac{1}{T} \sum_{t=1}^T (g_{i,d}^{(t)} - \bar{g}_{i,d})^2} = \mathbb{E}_d[\text{std}_t(g_{i,d})]. \quad (1)$$

High VoG marks examples whose input-gradient is unstable across training.

Data and models. Imagenette has 9,469 training and 3,925 validation images; we resize to 256, center-crop to 224, and apply ImageNet normalisation. We use ResNet-50 [He et al., 2016] and ConvNeXt-Base [Liu et al., 2022], each in two regimes: *pretrained* (ImageNet weights) and *from scratch* (random init). Downstream classifiers are trained with Adam (lr = 10^{-3} , weight decay 10^{-4}), cosine-annealed learning rate, gradient clipping at 1.0, and mixed precision, in two modes: *linear probe* (frozen backbone, trained head; pretrained only) and *fine-tuning* (all weights; both regimes). We report best validation accuracy over 10 epochs (pretrained) or 20 epochs (from scratch); all runs use a single seed (42), a limitation we return to in the Discussion. Subsets use the top-30% (High-VoG) and bottom-30% (Low-VoG) of examples. Full hyperparameters are in Appendix A.2.

Cross-architecture agreement. We compute VoG independently for ResNet-50 and ConvNeXt-Base and compare the two score vectors with Spearman rank correlation, Pearson correlation, the overlap of their top-30% sets, and per-class Spearman correlation. Table 1 reports the result. Under pretraining the rankings agree strongly ($\rho = 0.56$, top-30% overlap 64.6%), and the agreement holds *inside every class* (per-class $\rho > 0.5$ for all 10 classes). From random initialisation the agreement largely disappears ($\rho = 0.27$, overlap 38.9%, no class with $\rho > 0.5$). Pretrained features thus induce a shared notion of difficulty that random initialisation does not.

Table 1: **Cross-architecture agreement of VoG rankings** between ResNet-50 and ConvNeXt-Base on Imagenette. Pretrained models agree strongly on which examples are hard; trained from scratch they do not.

Initialisation	Spearman ρ	Pearson r	Top-30% overlap	Classes with $\rho > 0.5$
ImageNet-pretrained	0.56	0.58	64.6%	10/10
From scratch	0.27	0.22	38.9%	0/10

Does selection by VoG help? We fine-tune / linear-probe ConvNeXt-Base on Full data and on the four 30% subsets (High/Low VoG, ranked by ResNet-50 and by ConvNeXt’s own scores). Table 2 reports validation accuracy. Three patterns stand out. First, *High* $\gg$ *Low* in every regime: under pretrained fine-tuning the gap is 95.6 vs 88.1 (7.5 points). Second, under pretraining the high-VoG subset *exceeds* full data (95.6 vs 93.7, +1.9), i.e. removing the 70% lowest-VoG examples improves fine-tuning—consistent with redundancy of easy examples [Sorscher et al., 2022]. Third, the ResNet-ranked and ConvNeXt-ranked high-VoG subsets perform almost identically (95.6 vs 95.2), so the ranking transfers across architectures. Linear-probe accuracy is saturated ($\sim 99.9\%$ for all pretrained subsets) and therefore uninformative for selection; the action is in fine-tuning. We corroborate the within-architecture effect on ResNet-50 alone: fine-tuning reaches 95.8% on high-VoG vs 88.5% on low-VoG (full 96.1%).

Table 2: **ConvNeXt-Base validation accuracy (%) by training subset.** Each subset is 30% of the data except Full. “(R50)”/“(CNX)” denote which architecture’s VoG produced the ranking. Best non-Full result per column in bold. Pretrained: high-VoG matches or beats Full and is robust to the ranking source; from scratch: more data wins, but High $\gg$ Low persists.

Training subset	Pretrained		From scratch
	Lin. probe	Fine-tune	Fine-tune
Full (100%)	99.97	93.71	42.19
High-VoG (R50)	99.97	95.59	35.57
High-VoG (CNX)	99.97	95.16	35.85
Low-VoG (R50)	99.90	88.08	17.86
Low-VoG (CNX)	99.92	87.72	18.01

Is the ranking ordered by informativeness? As a finer check we split the data into ten disjoint VoG deciles and train one model per decile (10% of data each, 5 epochs). For pretrained models, test error tends to fall as VoG increases: ResNet-50 drops from 15.2% (lowest decile) to 6.9% (highest), and ConvNeXt-Base from 9.1% to 2.4% (Fig. 2). Higher-VoG examples are individually more useful for training, not less—difficulty here means informativeness.

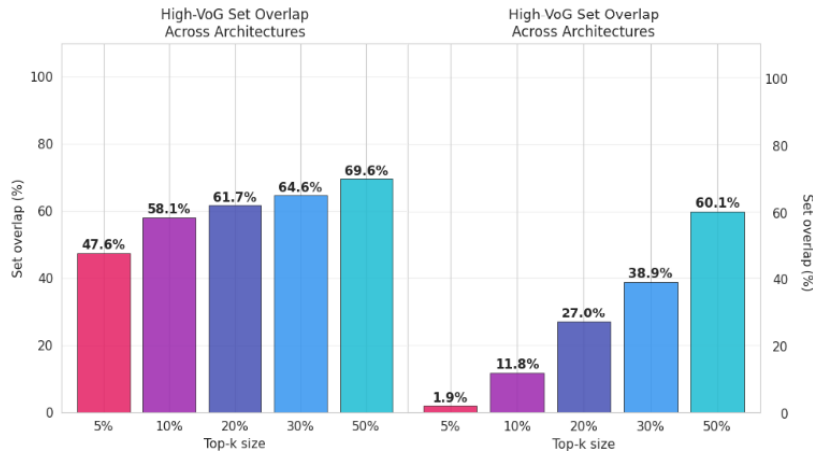


Figure 1: **Pretraining makes VoG difficulty agree across architectures.** Overlap (%) between the top- k highest-VoG examples selected by ResNet-50 and by ConvNeXt-Base, for $k \in \{5, 10, 20, 30, 50\}\%$. *Left*: both networks ImageNet-pretrained—the two “hard” sets overlap substantially already at $k=5\%$ (47.6%) and reach 69.6% at $k=50\%$. *Right*: both trained from scratch—overlap starts near chance (1.9% at $k=5\%$) and only approaches the pretrained level for large k . Takeaway: the two architectures agree on which examples are hard only when they share pretrained representations.

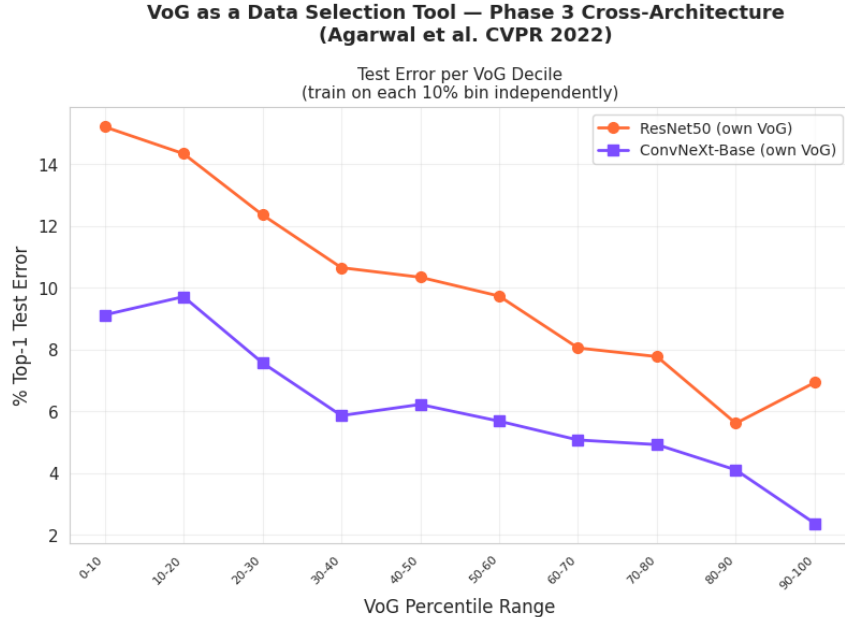


Figure 2: **Higher-VoG examples are more informative.** Top-1 test error of ResNet-50 (orange) and ConvNeXt-Base (purple), each trained on a single 10% VoG decile (5 epochs, both ranked by their own pretrained VoG). Error generally decreases as VoG increases, with small non-monotonic fluctuations (ResNet-50 15.2 $\rightarrow$ 6.9, ConvNeXt 9.1 $\rightarrow$ 2.4), so the low-VoG (easy) deciles carry the least training signal.

3 Discussion

What the results mean. VoG behaves as a property of the *data-through-pretrained-features*, not of the architecture. Once two networks are initialised from ImageNet, they agree on which Imagenette examples are hard (Table 1) and the high-VoG examples are interchangeably useful for either (Table 2). This is the mechanism that makes proxy-based selection [Coleman et al., 2020] work: a ResNet ranking is good enough to select data for a ConvNeXt. The decile analysis (Fig. 2) explains *why* pruning helps—the easy, low-VoG deciles contribute the least and can be dropped, echoing the prunability reported by Sorscher et al. [2022].

Where it fails. The transfer is not a property of VoG alone but of the shared representation. From random initialisation the agreement is weak (Table 1) and pruning no longer helps: every 30% subset underperforms full data (Table 2, right), because in the low-accuracy from-scratch regime the binding constraint is the amount of data, not its redundancy. VoG still orders informativeness correctly there (high $\gg$ low), but selection cannot substitute for scale. Our evidence is also limited: a single dataset, two architectures, and a single seed—the accuracy gaps in Table 2 are reported without variance and should be read as directional rather than precise.

What to do next. Two directions follow. First, we can repeat the cross-architecture comparison across seeds and at intermediate pretraining strengths (e.g. few-epoch warm starts) to map *how much* pretraining is needed before difficulty becomes shared. Second, we may test whether the pretrained ranking transfers to architectures outside the CNN family (e.g. ViT), which would distinguish “shared because both are ImageNet-pretrained” from “shared because both are convolutional.”

A Appendix

A.1 Additional quantitative results

Table 3 gives the full per-decile and cumulative top- $N\%$ test error underlying Fig. 2 (pretrained models, 5 epochs per run). The cumulative columns show that keeping only the highest-VoG examples holds error roughly constant down to $\sim 30\%$ of the data before degrading. Table 4 reports the single-architecture confirmation on ResNet-50.

Table 3: Per-decile and cumulative top- $N\%$ Top-1 test error (%), pretrained, 5 epochs per run. Deciles are disjoint 10% bins ordered by increasing VoG; cumulative keeps the top- $N\%$ highest-VoG examples.

Decile	Per decile		$N\%$	Cumulative top- $N\%$	
	R50	CNX		R50	CNX
0–10	15.2	9.1	100	4.4	6.3
10–20	14.3	9.7	90	4.4	5.2
20–30	12.4	7.6	80	4.9	5.3
30–40	10.7	5.9	70	4.2	5.1
40–50	10.3	6.2	60	4.6	4.6
50–60	9.7	5.7	50	4.0	3.8
60–70	8.1	5.1	40	4.2	4.5
70–80	7.8	4.9	30	4.4	3.9
80–90	5.6	4.1	20	4.5	3.5
90–100	6.9	2.4	10	6.9	2.5

Table 4: Phase 2 single-architecture check: ResNet-50 (pretrained) validation accuracy (%) by subset. High-VoG nearly matches Full; Low-VoG is markedly worse, mirroring the cross-architecture result.

Mode	Full	High-VoG (30%)	Low-VoG (30%)
Linear probe	99.36	99.29	96.56
Fine-tuning	96.10	95.80	88.46

A.2 Hyperparameters

Table 5 lists all settings needed to reproduce the experiments.

Table 5: Full experimental configuration. VoG follows Agarwal et al. [2022]; downstream training is identical across subsets.

Component	Setting
<i>Data</i>	
Dataset	Imagenette (10-class ImageNet subset)
Split	9,469 train / 3,925 val
Preprocessing	resize 256, center-crop 224, ImageNet mean/std
<i>Architectures</i>	
Models	ResNet-50; ConvNeXt-Base
Init regimes	ImageNet-pretrained; random (<code>weights=None</code>)
<i>VoG computation</i>	
Warm-up epochs T	5
Optimiser	SGD, lr 0.001, momentum 0.9, wd 10^{-4}
Batch size	32
Gradient target	true-class softmax probability
Model mode at snapshot	<code>eval()</code>
Gradient pooling	adaptive average-pool to 32×32
<i>Downstream training</i>	
Optimiser	Adam, lr 10^{-3} , wd 10^{-4}
LR schedule	cosine annealing
Gradient clipping	max-norm 1.0
Precision	mixed (AMP)
Loss	cross-entropy
Batch size	128 (Phase 2) / 64 (Phase 3)
Epochs	10 (pretrained) / 20 (from scratch)
Modes	linear probe (frozen) / fine-tuning (all weights)
<i>Selection & evaluation</i>	
Subset fraction	30% (High-VoG / Low-VoG)
Decile runs	10 disjoint 10% bins, 5 epochs each
Cumulative runs	top- $N\%$, $N \in \{100, \dots, 10\}$, 5 epochs each
Seed	42 (single run)
Hardware	2× NVIDIA T4 (single-GPU training)

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